

Simple Linear Regression

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Regression

Regression is the broad idea, and simple linear regression, multiple linear regression, and logistic regression are different types/models.

Simple linear regression only has one predictor.

Data and plot

```
library(ggplot2)
```

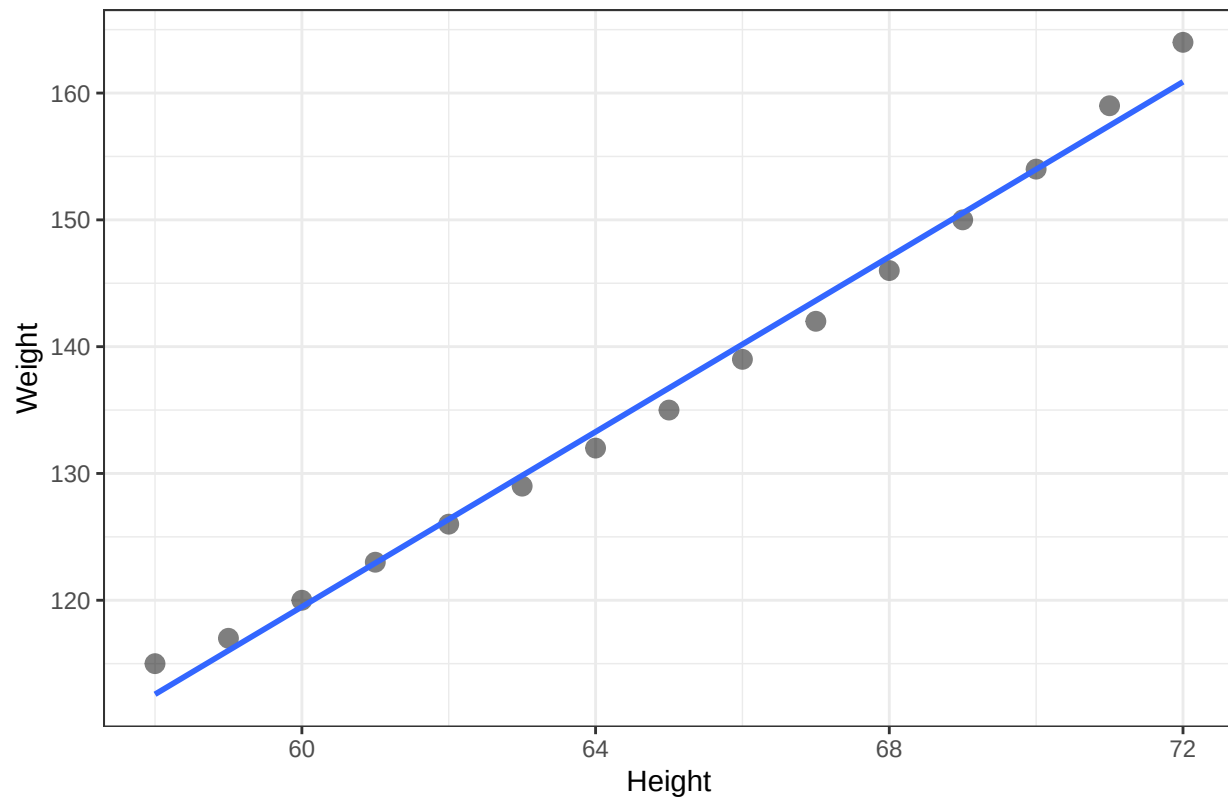
```
head(women)
```

```
##   height weight
## 1     58    115
## 2     59    117
## 3     60    120
## 4     61    123
## 5     62    126
## 6     63    129
```

```
women |>
  ggplot(aes(height, weight)) +
  geom_point(size = 3, alpha = .5) +
  geom_smooth(method = lm, se = FALSE) +
  theme_bw() +
  labs(
    title = "Weight explained by Height in Women",
    x = "Height",
    y = "Weight"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Weight explained by Height in Women



Linear model

A linear model asks: How much of the change in weight can be explained by change in height?

```
model1 <- lm(weight ~ height, data = women)
```

```
summary(model1)
```

```
##
## Call:
## lm(formula = weight ~ height, data = women)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7333 -1.1333 -0.3833  0.7417  3.1167
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -87.51667    5.93694  -14.74 1.71e-09 ***
## height       3.45000    0.09114   37.85 1.09e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.525 on 13 degrees of freedom
```

```
## Multiple R-squared:  0.991,  Adjusted R-squared:  0.9903
## F-statistic: 1433 on 1 and 13 DF,  p-value: 1.091e-14
```

The slope is approximately 3.45. This means that for every one-unit increase in height, the predicted weight increases by approximately 3.45 units.

The R-squared is approximately 0.991, meaning that about 99.1% of the variation in weight is explained by height in this linear model.